LETTER TO THE EDITOR

APOE3ch alleviates Aβ and tau pathology and neurodegeneration in the human APP^{NL-G-F} cerebral organoid model of Alzheimer's disease

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Dear Editor,

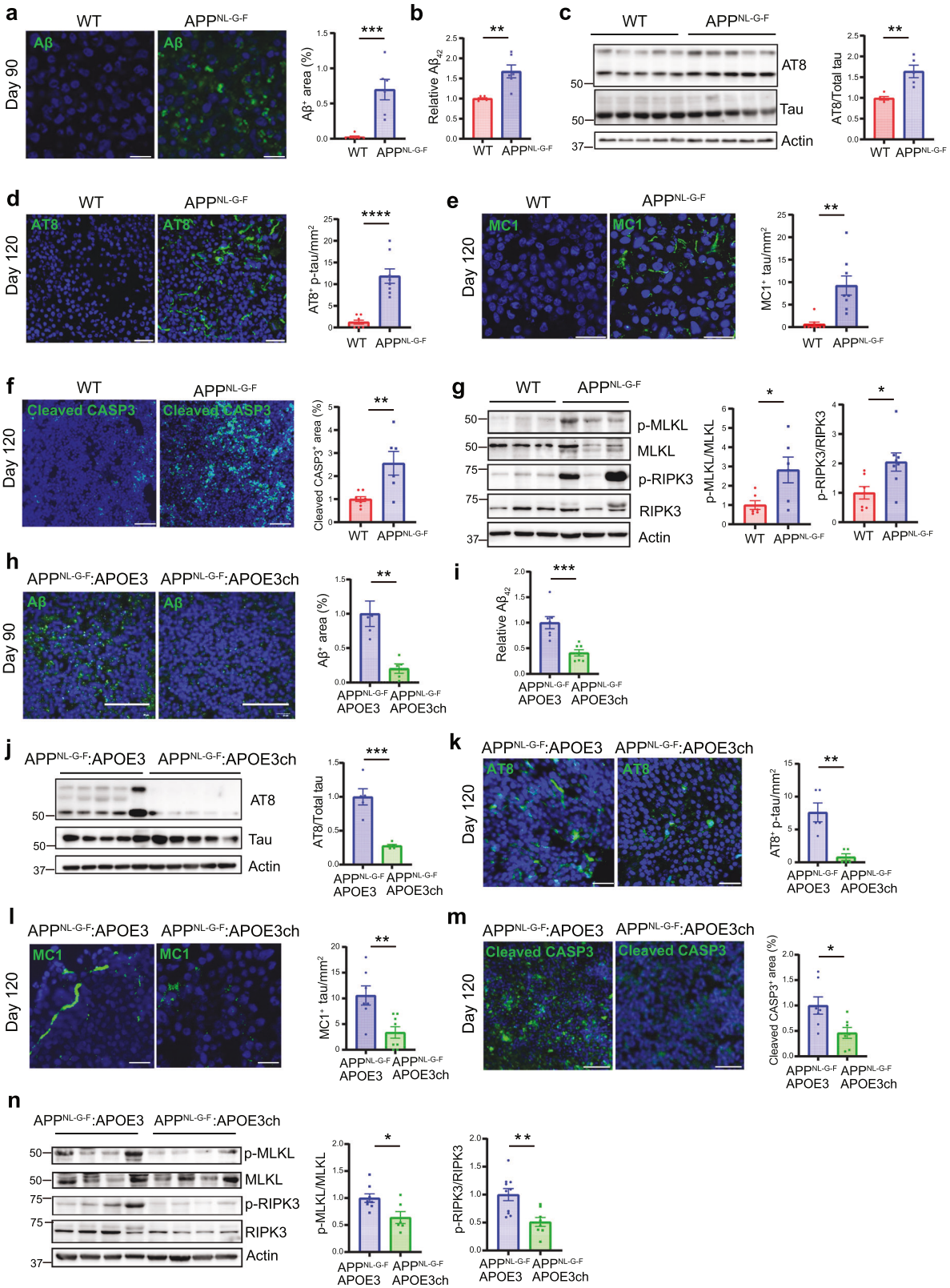
Research in Alzheimer's disease (AD) has greatly benefited from the genetic studies of rare autosomal dominant mutations.[1] In a recent case study of the world's largest early onset AD pedigree with presenilin 1 (*PSEN1*) mutation, an individual was identified to have a high amyloid burden but low tau measurements, and an unusually delayed onset of cognitive impairment.[2] Whole-exome sequencing revealed an additional rare homozygous mutation of *APOE3* Christchurch (*APOE3ch,* R136S) as the potential protective factor for AD. *APOE3* is one of variants in the Apolipoprotein E (*APOE*) gene, which has been extensively studied for its association with sporadic AD.[3] *APOE* has three variants, ε2, ε3 and ε4; and *APOE* ε4 (*APOE4*) allele confers significantly increased risk for developing AD, relative to the common *APOE* ε3 (*APOE3*) allele. Exactly how different variants of APOE alter amyloid-β (Aβ) and tau pathology and neurodegeneration is an active area of research.[4] While the identification of *APOE3ch* mutation opens a door for new investigations, the rarity of the APOE3ch makes it hard to study its underlying mechanisms in humans. Further, whether the *APOE3ch* mutation affects neuronal death and other AD pathophysiology such as neuroinflammation remains to be demonstrated.

To examine the biological effects of APOE3ch in AD, we have developed a human cerebral organoid model of AD. Although few rodent models could fully recapitulate AD pathophysiology,[5] we previously showed that knocking in the Swedish (KM670/671 NL), Arctic (E693G) and Beyreuther/Iberian (I716F) mutations of amyloid precursor protein (*APP*) into the rat genome leads to a rat AD model (*App*$^{NL-G-F}$) with pathologies and disease progression resembling those in AD patients.[6] This prompted us to develop a human AD organoid model, using a similar strategy. We introduced the three familial *APP* mutations into human H1 embryonic stem cells (hESCs), using CRISPR-Cas9 gene editing (Supplementary information, Fig. S1a). DNA sequencing validated the homozygous mutation of *APP*$^{NL-G-F}$ in hESCs (Supplementary information, Fig. S1b). The wild-type (WT) and APP^{NL-G-F} hESCs were induced to differentiate into forebrain organoids.[7] After 45 days of development in culture, the organoids exhibited cortex-like structure with cells expressing SOX2, a marker for neural progenitor cells in the cortical plate, and those expressing CTIP2, a marker for neurons with a deep cortical layer identity (Supplementary information, Fig. S2a). Later, at day 60, the organoids extensively expressed MAP2, a marker for differentiated neurons (Supplementary information, Fig. S2b). Further, cells expressing the astrocyte marker GFAP were found to emerge among many NeuN$^+$ neuron-like cells (Supplementary information, Fig. S2c). There was no obvious difference between WT and APP^{NL-G-F} organoids in the speed of development and differentiation into neurons and astrocytes. Few microglia were seen in the cerebral organoids. This is expected in that microglia are derived from myeloid cells in the mesoderm, whereas neurons and astrocytes are derived from the ectoderm.[8]

In human AD brain, Aβ deposition, tau pathology and neurodegeneration occur sequentially.[9] We therefore examined these AD pathologies in the APP^{NL-G-F} forebrain organoids at different timepoints. We observed that in the APP^{NL-G-F} organoids but not in WT organoids, Aβ deposition appeared at day 60 as shown by immunofluorescence using D54D2 anti-Aβ antibody (Supplementary information, Fig. S3a), and gradually accumulated to form comprehensive Aβ aggregates at day 90 (Fig. 1a). Consistently, quantitative analysis by ELISA indicated a significant increase in secreted Aβ42 from the medium of APP^{NL-G-F} but not WT neurons (Fig. 1b). Next, we examined tau pathology, including hyper-phosphorylated tau (p-tau) using the AT8 antibody and misfolded tau using the MC1 antibody. Western blotting showed a modest but significantly increased level of AT8$^+$ p-tau at Ser202/Thr205 in the APP^{NL-G-F} organoids at day 90 (Fig. 1c). Immunofluorescence staining revealed a marked increase of AT8$^+$ p-tau immunoreactivity in the 120-day APP^{NL-G-F} organoids (Fig. 1d). Further, we determined misfolded tau with conformational changes (MC1$^+$).[10] Immunofluorescence staining indicated the presence of MC1$^+$ tau aggregates in the 120-day APP^{NL-G-F} organoids. Comparatively, no immunoreactivity for MC1 was found in the WT organoids (Fig. 1e). Given that gliosis develops in response to Aβ deposition and tau pathology,[11] we examined the pathological states of astrocytes. Astrocytes labeled with GFAP or S100β indicated astrogliosis in the APP^{NL-G-F} organoids (Supplementary information, Fig. S4a, b). Further, western blotting showed significantly increased GFAP levels in the APP^{NL-G-F} organoids at day 90, as compared with the WT controls (Supplementary information, Fig. S4c).

In addition to the notable AD pathology, neurodegeneration was also evident in the APP^{NL-G-F} organoids. There was evident decrease in MAP2 immunoreactivity in 90-day APP^{NL-G-F} organoids, compared with their WT counterpart, suggesting neuron damage (Supplementary information, Fig. S5a). We thus determined whether apoptosis and necroptosis underlie neurodegeneration.[12] Notable levels of cleaved caspase-3 as revealed by immunofluorescence indicated apoptosis in the APP^{NL-G-F} organoids at day 120, whereas significant lower levels of cleaved caspase-3 were observed in the WT controls (Fig. 1f). Meanwhile, increased MLKL and RIPK3 phosphorylation (p-MLKL/MLKL and p-RIPK3/RIPK3) corroborated the presence of necroptosis in the APP^{NL-G-F} organoids (Fig. 1g). Taken together, these results illustrate sequential development of Aβ aggregates, tau pathology, astrogliosis and cell death in the APP^{NL-G-F} cerebral organoids,

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resembling the full spectrum of pathological and pathophysiological events in human AD brain (Supplementary information, Fig. S5b).

The generation of the human APP^{NL-G-F} cerebral organoid model allowed us to determine the potential protective effects of APOE3ch against AD pathologies and neuronal death. We used CRISPR-Cas9 gene editing to knock in *APOE3ch* into the genome of APP^{NL-G-F} hESCs, which expressed *APOE3* allele (Supplementary information, Fig. S6a, b). We then proceeded to establish forebrain organoids expressing normal APOE3 (APP^{NL-G-F}:APOE3) or

Fig. 1 APOE3ch alleviates Aβ and tau pathology and suppresses neurodegeneration in the APP^{NL-G-F} forebrain organoids. a Significant Aβ aggregates observed in the APP^{NL-G-F} organoids. Representative images (left) and quantification (right) of the Aβ puncta in the 90-day organoids as revealed by immunofluorescence using D54D2 anti-Aβ antibody ($n = 6$ biologically independent organoids in each genotype). Scale bars, 10 μm. **b** ELISA analyses showing a marked increase of Aβ42 levels in the medium of APP^{NL-G-F} neurons, relative to WT ($n = 6$ supernatants from each genotype). **c** Significant increase in p-tau at Ser202/Thr205 in the APP^{NL-G-F} organoids. Representative western blots (left) showing an increase in the level of AT8$^+$ p-tau and quantification (right) of AT8$^+$ p-tau/total tau ratios in the APP^{NL-G-F} organoids at day 90, as fraction of WT organoids ($n = 5$ organoids in each genotype). **d** Immunostaining of AT8$^+$ p-tau immunoreactivity at day 120 ($n = 8$ organoids in each genotype). Number of AT8$^+$ p-tau/mm^2 was shown. Scale bars, 50 μm. **e** Appearance of misfolded tau in the APP^{NL-G-F} organoids at day 120. Representative images (left) and quantification (right) of MC1$^+$ misfolded tau in the 120-day organoids ($n = 8$ organoids in each genotype). Number of misfolded tau/mm^2 was quantified. Scale bars, 20 μm. **f** Significant apoptosis of neurons in the APP^{NL-G-F} organoids at day 120. Apoptotic cells are shown by immunostaining of cleaved caspase-3 ($n = 8$ for WT organoids, $n = 6$ for APP^{NL-G-F} organoids). Scale bars, 80 μm. **g** Presence of necroptosis in the APP^{NL-G-F} organoids. Western blots (left) and quantification (right) of p-MLKL/MLKL and p-RIPK3/RIPK3 ratios in the APP^{NL-G-F} organoids at day 90, as fraction of WT (2 organoids mixed in one sample; for p-MLKL/MLKL: $n = 6$ samples for WT, $n = 6$ samples for APP^{NL-G-F}; for p-RIPK3/RIPK3: $n = 7$ samples for WT, $n = 8$ samples for APP^{NL-G-F}. **h** Expression of APOE3ch reduces Aβ aggregates in the APP^{NL-G-F} organoids. Representative immunofluorescence images (left) and quantification (right) of Aβ puncta showing a marked reduction in Aβ puncta in the APP^{NL-G-F}:APOE3ch organoids, compared with those in APP^{NL-G-F}:APOE3 organoids, at day 90 ($n = 5$ organoids in each genotype). Scale bars, 80 μm. **i** ELISA analyses showing a significant decrease of $A\beta_{42}$ levels in culture medium from APP^{NL-G-F}:APOE3ch neurons, compared with that from APP^{NL-G-F}:APOE3 ($n = 7$ media from each genotype). **j** Expression of APOE3ch inhibits tau phosphorylation. Western blotting analysis of AT8$^+$ p-tau and quantification of AT8$^+$ p-tau/total tau ratio showing a marked decrease in p-tau in the APP^{NL-G-F}:APOE3ch organoids compared with APP^{NL-G-F}:APOE3 organoids at day 90 (2 organoids mixed in one sample, $n = 5$ biologically independent samples for each genotype). **k** Immunostaining and quantification of AT8$^+$ p-tau in the APP^{NL-G-F}:APOE3 and APP^{NL-G-F}:APOE3ch organoids at day 120. Note that the number of AT8$^+$ p-tau was decreased after expressing APOE3ch ($n = 5$ organoids in each genotype). Number of AT8$^+$ p-tau/mm^2 was calculated. Scale bars, 40 μm. **l** Expression of APOE3ch reduces misfolded tau. Representative images (left) and quantification (right) of MC1$^+$ misfolded tau in the APP^{NL-G-F}:APOE3 and APP^{NL-G-F}:APOE3ch organoids at day 120 ($n = 7$ for APP^{NL-G-F}:APOE3 organoids, $n = 8$ for APP^{NL-G-F}:APOE3ch organoids). Number of misfolded tau/mm^2 was suggested. Scale bars, 20 μm. **m** APOE3ch protects against apoptosis as revealed by immunostaining of cleaved caspase-3 in the APP^{NL-G-F}:APOE3 and APP^{NL-G-F}:APOE3ch organoids at day 120 ($n = 7$ organoids for each genotype). Scale bars, 50 μm. **n** APOE3ch inhibits necroptosis as revealed by western blotting analysis of the necroptosis markers of p-MLKL/MLKL and p-RIPK3/RIPK3 in the APP^{NL-G-F}:APOE3 and APP^{NL-G-F}:APOE3ch organoids (2 organoids mixed in one sample; for p-MLKL/MLKL: $n = 8$ samples for APP^{NL-G-F}:APOE3, $n = 6$ samples for APP^{NL-G-F}:APOE3ch; for p-RIPK3/RIPK3: $n = 10$ samples for APP^{NL-G-F}:APOE3, $n = 8$ samples for APP^{NL-G-F}:APOE3ch). Data are shown as means ± SEM. *$P < 0.05$, **$P < 0.01$, ***$P < 0.001$, ****$P < 0.0001$, unpaired two-tailed Student's *t*-test.

APOE3ch (APP^{NL-G-F}:APOE3ch), respectively. We did not find a significant difference in APOE3 expression in organoids harboring *APOE3* or *APOE3ch* (Supplementary information, Fig. S6c). Remarkably, expression of APOE3ch significantly reduced Aβ deposition, as revealed by immunostaining using D54D2 anti-Aβ antibody (Fig. 1h, left). Quantification of Aβ$^+$ immunofluorescence images showed a 5-fold decrease in the Aβ$^+$ puncta in sections at day 90 in the APP^{NL-G-F}:APOE3ch organoids, as compared to those in the control APP^{NL-G-F}:APOE3 organoids (Fig. 1h, right). Consistently, ELISA analyses showed a marked decrease in the levels of secreted Aβ42 in the medium from APP^{NL-G-F}:APOE3ch neurons (Fig. 1i). Next, we determined whether APOE3ch confers protection against tau pathology. Western blotting of p-tau using AT8 antibody showed a significant lower tau phosphorylation at Ser202/Thr205 in the APP^{NL-G-F}:APOE3ch organoids than that in the APP^{NL-G-F}:APOE3 organoids (Fig. 1j). Immunofluorescence staining experiments also indicated a significant decrease in the AT8$^+$ p-tau immunoreactivity in the organoids expressing APOEch (Fig. 1k). Moreover, misfolded tau with conformational changes was notably reduced by expression of APOEch, as revealed by immunostaining by MC1 anti-tau antibody (Fig. 1l). We next determined less astrogliosis is associated with mitigated Aβ and tau pathology, as indicated by immunofluorescence of GFAP or S100β (Supplementary information, Fig. S7a, b). Quantitative analyses by western blotting showed reduced GFAP expression (Supplementary information, Fig. S7c). Finally, we examined the effects of APOEch on neurodegeneration. Remarkably, expression of APOE3ch significantly increased levels of MAP2, suggesting enhanced neuron survival (Supplementary information, Fig. S8a). Immunostaining using antibody against cleaved caspase-3 showed a marked reduction in the cleaved caspase-3 immunoreactivity in APP^{NL-G-F}:APOE3ch organoids compared with that in APP^{NL-G-F}:APOE3 organoids, suggesting APOE3ch expression inversely correlated with apoptosis (Fig. 1m). Moreover, Western blotting assay demonstrated a significant reduction in the ratio of p-MLKL to total MLKL and p-RIPK3 to RIPK3, indicating a protective role of APOE3ch against necroptosis (Fig. 1n). Taken together, we demonstrated that APOE3ch expression potently alleviates Aβ deposition, tau phosphorylation and aggregation, astrogliosis and cell death in human APP^{NL-G-F} cerebral organoids.

There are two important contributions of the present study. First, we have developed a human APP^{NL-G-F} cerebral organoid model of AD, which resembles the continuum of AD pathologies and neurodegeneration seen in human AD brains. Compared with more than 200 animal models for AD, this AD cerebral organoid model is fully human and develops key pathological features at a relatively fast pace (2–3 months vs 6–12 months in animals).[5] Further, in comparison with the other AD organoid models, mostly derived from human induced pluripotent stem cells generated from carriers of *PSEN1*, *PSEN2*, *APP* pathogenic mutations or *APOE4*,[13] our hESCs-derived APP^{NL-G-F} organoids by gene editing show superiority in terms of homogeneous genetic background. Thus, this model may facilitate mechanistic studies of AD and drug testing. Second, we have demonstrated the protective effects of APOE3ch against AD pathology and neurodegeneration in human cerebral organoids. Consistent with the case report of *APOE3ch* carrier in the autosomal dominant AD pedigree, expression of APOE3ch in human APP^{NL-G-F} organoids potently reduced tau pathology. We also found that APOE3ch expression alleviates Aβ deposition, astrogliosis, and cell death. Exactly how APOE3ch elicits its protection against multiple AD pathophysiological events remains unknown. Our APP^{NL-G-F} organoid model offers a unique opportunity to effectively investigate the unique features of APOE3ch and delineate the cellular and molecular mechanisms underlying its resistance to AD.

During the preparation of this manuscript, two papers on APOE3ch appeared. In one study, Chen and colleagues generated a humanized *APOE3ch* knock-in mouse line and crossed it with Aβ-based APP/PS1 mice.[14] They found that APOE3ch alleviates X34$^+$ fibrillar Aβ plaques but not Aβ oligomers, and Aβ-associated tau pathology, which was attributed to increased myeloid cell phagocytosis. In the other study, Nelson et al. knocked in *APOE3ch* into the APOE4 mice and crossed them with the tau transgenic mice PS19.[15] They showed that homozygous APOE3ch rescued

APOE4-driven tau pathology, hippocampal volume and gliosis. Although these studies demonstrated the same protective role of APOE3ch against AD pathology as ours, a number of important differences are noted. An obvious difference was that we used fully human cerebral organoid system. In addition, the APP^{NL-G-F} organoid system is based on knock-in but not transgenic overexpression of pathogenic mutations, and therefore avoids potential non-physiological phenotypes due to expression of the transgenes. Finally, the APP^{NL-G-F} cerebral organoids develop both Aβ and tau pathology, astrogliosis as well as apoptosis and necroptosis, which are prominent in AD human brains but difficult to be recapitulated in APP/PS1 mice or PS19 transgenic mice. These features of the APP^{NL-G-F} organoids allowed us to demonstrate the remarkable effects APOE3ch in alleviating neuronal apoptosis and necroptosis. Taken together, we believe this paper reports a new human organoid model urgently needed in the AD research field, and provides a comprehensive demonstration of the protective effect of APOE3ch against AD pathology and neurodegeneration. Given minimum microgliosis in the AD organoids, we speculate that APOE3ch may inhibit Aβ- and/or tau-dependent cell death through a cell autonomous mechanism, and microglia may elicit regulatory effects on this process. Further investigation may delineate whether microglia regulate neuronal death directly or through alteration of Aβ or tau pathology.

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AUTHOR CONTRIBUTIONS

B.L., H.L. and F.M. conceived the study and designed the major experiments. H.L., F.M., R.Y., X.H., S.W., Y.D., Y.Z., K.P. and W.G. performed the experiments and analyzed the data. The manuscript was written by B.L. and F.M. with input from all of the authors.

COMPETING INTERESTS

The authors declare no competing interests.

ADDITIONAL INFORMATION

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